

VERKHOTURYE, Russian Federation

WMO: 281440

Lat: 58.867N

Lon: 60.783E

Elev: 412

StdP: 14.48

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.2	-24.0	-34.5	0.8	-28.9	-29.1	1.1	-23.9	14.4	13.1	12.7	12.7	2.4	20	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	84.5	68.1	81.3	66.2	77.9	64.2	70.0	81.4	68.1	78.8	66.2	75.1	5.0	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	97.3	75.0	64.1	90.9	73.1	62.4	85.4	71.0	34.2	81.8	32.6	78.6	31.1	75.2	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	15.3	12.9	11.4	DB	-36.1	89.3	4.8	2.6	-39.5	91.2	-42.3	92.7	-45.0	94.2	-48.5	96.1
(5)				WB	-35.7	72.1	5.0	2.2	-39.3	73.7	-42.3	75.0	-45.1	76.2	-48.8	77.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	34.7	3.7	7.4	23.1	37.2	50.1	59.7	63.4	58.4	48.1	36.3	18.6	8.4
(7)		DBStd	23.54	13.89	14.31	10.19	9.44	9.25	8.13	7.13	7.42	7.99	8.96	14.59	15.17
(8)		HDD50	6799	1436	1194	835	393	117	15	3	10	129	435	943	1289
(9)		HDD65	11207	1901	1614	1300	833	467	198	117	230	509	891	1393	1754
(10)		CDD50	1206	0	0	0	10	121	306	419	271	71	8	0	0
(11)		CDD65	138	0	0	0	0	5	38	67	27	1	0	0	0
(12)		CDH74	1667	0	0	0	1	139	459	762	290	16	0	0	0
(13)		CDH80	410	0	0	0	0	15	105	198	91	1	0	0	0
(14)	Wind	WSAvg	4.9	4.3	4.7	5.6	5.6	5.1	4.3	4.3	4.6	5.3	5.0	4.6	
(15)	Precipitation	PrecAvg	22.9	1.1	0.7	1.0	1.1	2.0	3.0	3.7	3.4	2.6	1.5	1.4	1.1
(16)		PrecMax	28.3	2.4	1.8	2.0	2.8	4.0	6.1	7.4	6.2	5.6	3.0	2.4	2.0
(17)		PrecMin	18.5	0.2	0.2	0.0	0.1	0.6	0.7	0.4	0.9	0.4	0.4	0.2	0.4
(18)		PrecStd	2.7	0.5	0.4	0.6	0.7	1.0	1.5	1.8	1.5	1.3	0.7	0.6	0.5
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.6	38.0	49.2	71.1	82.0	86.4	88.8	87.8	77.2	64.5	45.8	37.3
(20)		0.4%	MCWB	33.6	33.6	39.1	53.6	61.1	67.4	69.9	69.2	60.6	53.5	42.3	34.8
(21)		2%	DB	31.1	34.4	43.7	62.6	78.4	83.1	85.0	81.7	70.9	58.6	42.1	34.8
(22)		2%	MCWB	29.2	31.5	36.1	49.3	59.8	66.4	69.2	67.7	58.5	50.0	39.1	32.2
(23)		5%	DB	27.1	31.4	40.1	57.4	73.6	79.7	81.8	76.0	64.9	53.7	38.0	32.1
(24)		5%	MCWB	25.5	28.5	33.8	46.2	56.9	64.8	68.2	64.8	56.0	47.5	35.5	29.9
(25)		10%	DB	23.1	27.8	36.9	53.0	68.6	75.7	78.3	71.9	60.4	49.2	34.8	28.5
(26)		10%	MCWB	21.5	25.1	32.1	43.2	54.7	62.7	65.9	62.6	54.2	44.8	32.7	26.4
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.4	34.5	41.1	54.7	63.9	70.9	72.8	70.9	63.0	54.0	43.0	35.1
(28)		0.4%	MCDWB	35.3	37.0	46.9	68.2	76.8	81.9	83.3	84.0	72.8	61.1	45.1	36.7
(29)		2%	WB	29.4	31.7	37.1	50.3	61.0	68.3	70.7	68.2	59.9	51.1	39.6	32.5
(30)		2%	MCDWB	31.0	34.6	42.4	61.6	74.8	79.7	81.9	79.5	67.5	57.3	42.0	34.7
(31)		5%	WB	25.7	28.6	34.6	46.8	58.6	66.0	69.1	66.0	57.2	47.9	35.8	30.0
(32)		5%	MCDWB	27.1	31.3	38.8	55.7	71.4	76.9	79.6	74.5	63.6	53.5	37.7	32.0
(33)		10%	WB	21.5	25.2	32.6	43.9	55.8	63.9	67.3	63.8	54.8	44.9	33.0	26.7
(34)		10%	MCDWB	23.0	27.7	36.5	51.9	66.7	74.0	76.7	70.4	59.9	49.1	34.7	28.4
(35)	Mean Daily Temperature Range	MDBR	13.7	17.9	18.3	18.4	21.9	19.5	19.8	17.5	14.5	11.8	11.2	12.6	
(36)		5% DB	MCDBR	12.6	14.6	20.0	27.0	30.4	26.0	26.0	25.3	21.0	19.6	10.3	9.5
(37)		5% DB	MCWBR	11.8	12.0	13.7	16.2	15.0	11.7	11.9	12.2	11.2	12.0	8.5	9.0
(38)		5% WB	MCDBR	12.5	13.9	16.4	24.9	27.4	23.8	23.9	23.1	18.3	18.4	9.9	10.1
(39)		5% WB	MCWBR	11.9	11.7	11.8	15.5	14.9	11.8	11.6	11.7	10.7	11.7	8.4	9.6
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.306	0.362	0.401	0.400	0.398	0.416	0.427	0.376	0.338	0.293	0.266	
(41)		taud	2.030	2.047	1.974	2.089	2.273	2.337	2.290	2.227	2.329	2.301	2.127	1.980	
(42)		Ebn at Noon	160	217	236	252	263	266	257	242	237	209	154	117	
(43)		Edh at Noon	17	27	40	43	38	36	37	37	29	22	15	12	
(44)	All-Sky Solar Radiation	RadAvg	159	440	843	1257	1578	1689	1691	1251	752	375	164	88	
(45)		RadStd	18	51	76	147	136	135	154	128	82	56	23	11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page